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**Commissioner,**

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# HEALTH AND HYGIENE

## INTRODUCTION

Good health is an asset for individuals, their communities and the nation at large. A nation cannot progress without a healthy population. There are large numbers of factors which affect our health. Some of these are balanced food, clean water and clean environment which help to remain healthy while others such as germs and stressful environment cause diseases and disorders.

**Definition of health:**

As defined by World Health Organization (WHO), it is a "State of complete physical, mental, and social wellbeing, and not merely the absence of disease or infirmity." Health is a dynamic condition resulting from a body's constant adjustment and adaptation in response to stresses and changes in the environment for maintaining an inner equilibrium called homeostasis.

**Hygiene** refers to the good habits and their practices which is followed to prevent diseases, maintain good health, especially through cleanliness, consumption of safe drinking water and proper disposal of sewage. It refers to all those activities that are done for improving and maintaining good health and sound mind.

**Signs of good health:**

**A. Signs of physical health:**

If you have good physical health you,

1. Shall be energetic and alert.
2. Have weight which is normal for your age and height.
3. Have bright and shining eyes.
4. Have all the body organs functioning normally and fall sick less often.
5. Have clean and clear skin.

6. Have normal growing hair of natural colour and texture.
7. Have odourless breath.
8. Have good appetite and
9. Get sound sleep.

Healthy individuals are active, responsive and happy and can work hard and perform well

**B. Signs of mental health:**

If you have good mental health, you shall have,

1. Control over your emotions
2. Balanced feelings, desires, ambitions and ideas
3. The ability to accept the realities of life and face them
4. Confidence in yourself
5. The ability to cope with the normal stresses of life
6. Sensitivity towards needs of others
7. The ability to give and seek help when needed
8. The ability to cope with conflict and disagreement

**C. Signs of social health:**

If you have good social health, you shall

1. Have a positive attitude towards life
2. Get along well with others
3. Have a pleasant personality
4. Fulfil responsibility/duties towards others
5. Have healthy interpersonal relationships
6. Be able to express disagreement positively

Such people form a progressive society that tries to address social issues and find solutions.

## Personal and Community Health

Both personal and community health are important for the wellbeing of an individual as well as the community. Health may be affected by the environment, availability of quality food and other necessities. Participation of local and government organizations becomes essential in promoting community health.

### Personal Health:

### (i) Balanced diet:

A balanced diet is one containing carbohydrates, proteins, fats, vitamins, minerals and fibres in correct proportion to meet the nutritional requirement of a person at a certain age and doing a certain amount of physical work. The energy requirement for moderately active adolescent (teenage) boy is about 2200 to 2400 Calories whereas for adolescent girls it is 2000 Calories.

### (ii) Personal hygiene:

Following are some necessary hygienic habits.

1. ***Regular toilet habit:*** Regular bowel movement keeps us free of waste generated within the body.
2. ***Washing hands before eating****:* Our hands may carry many disease-causing germs and therefore we must wash them with soap or ash before taking food.
3. ***Bathing regularly and wearing clean clothes:*** Bathing regularly keeps our body free of dirt, body lice and germs.
4. ***Cleaning the teeth:*** When we eat, some food particles may remain stuck to our teeth. These particles attract germs which harm our gums and teeth, and cause bad breath. Brushing teeth every morning and after dinner helps in their removal.
5. ***Washing hair, cleaning eyes, ears and nails:*** Regular washing and combing of hair, and washing eyes with clean water helps to keep the germs away. Nails of both hands should be clipped regularly. Nail biting is unhygienic and must be avoided.

**(iii) Domestic hygiene:**

We dust and mop our houses to keep them clean and free from dirt, flies and germs. Disposal of garbage in the allotted bins ensures clean surroundings and prevents breeding of disease-causing organisms. Used sanitary napkins need to be wrapped and put in the garbage bins. All utensils that are used at home should be kept clean.

**(iv) Clean food and water:**

Before they are consumed, fruits and vegetables should be washed thoroughly in clean water to make them free from germs.

**(v) Cooking with care**

1. Food should be prepared in a clean kitchen.
2. While cooking food, it is important to heat it to high temperature to kill germs.
3. Cooked food should be eaten fresh or covered and stored in cool and clean place. If stored, food should be kept at a low temperature using either conventional methods or in a refrigerator.

**(vi) Exercise**

Regular walking, yoga and physical exercises and outdoor games keep the heart and circulatory system in good condition. Walking keeps the joints healthy.

**(vii) Regular sleep and relaxation**

Is necessary for good health.

**(viii) Abstaining from habit-forming:**

Its substances such a smoking, chewing of betel nut, gutka, tobacco, and drinking alcohol;

These are addictive and their continuous use damages the liver, kidney, heart, gums and teeth. You may have noticed that spitting is quite common among people who chew betel nut, gutka and tobacco. We must stay away from such habits and try to convince others to do the same.

**Community health:**

Community health is the practice of preventing disease and promoting health of a population through the organized efforts of society, public and private organizations, communities and individuals. There are various activities and programmes undertaken by the government and local organizations to maintain health of the people and keep them free from disease.

Some of these activities and programmes are listed below:

1. Maintain cleanliness of the villages, towns, cities and localities through proper garbage disposal.
2. Ensure that prescribed standards in food stores, meat and milk outlets are strictly followed.
3. Organize health promotion and disease prevention, including
   a) Immunization programmes for infants and children, for instance The National Pulse Polio Programme that is being run by the government.
   b) Various awareness programmes have been undertaken against the spread of diseases of national concerns like malaria, AIDS, polio, leprosy, tuberculosis and hepatitis B. The National Malaria Eradication programme is an example of one such programme.
4. Organize mid-day meals in schools to provide adequate nourishment to growing children.
5. Set up hospitals and dispensaries to provide medical facilities to the general public either free of cost or at highly subsidized rates.

   To keep the environment clean we need to:

   1. Keep our house clean.
   2. Keep our neighborhood clean.
   3. An effective drainage system is essential as stagnated water allows breeding of mosquitoes.

**Human diseases:**

Disease can be defined as a disorder or malfunction of the mind or body. It involves morphological, physiological and psychological disturbances which may be due to environmental factors or pathogens or genetic anomalies or life style changes.

Diseases can be broadly grouped into infectious and non-infectious types. Diseases which are transmitted from one person to another are called infectious diseases or **communicable diseases**. Such disease-causing organisms are called **pathogens** and are transmitted through air, water, food, physical contact and vectors. The disease-causing pathogen may be virus, bacteria, fungi, protozoan parasites, helminthic parasites, etc.,

**Infectious diseases** are common and everyone suffers from such diseases at some time or the other. Most of the bacterial diseases are curable but all viral diseases are not. Some infectious disease like AIDS may be fatal.

**Non-infectious diseases** are not transmitted from an infected person to a healthy person. In origin they may be genetic (cystic fibrosis), nutritional (vitamin deficiency diseases) and degenerative (arthritis, heart attack, stroke). Among non - infectious diseases, cancer is one of the major causes of death.

**Kwashiorkor:**

Kwashiorkor is acute malnutrition that occurs mostly in children. This condition results from severe protein deficiencies. It is also known as “edematous malnutrition” because it is associated with edema. The patients suffering from Kwashiorkor have an emaciated appearance all over the body. The ankles feet and belly swell with liquid.

Kwashiorkor is most common in rural areas, especially in the sub-Saharan regions. The famine struck areas or the areas with low food supply are more prone to this disease. Also, the areas where people are unaware of proper diet and nutrition have more cases of Kwashiorkor.

**Causes:**

Protein is required by our body to repair the cells and make new cells. It is an important nutrient during pregnancy and a child's growth. Lack of proteins shuts down the growth and normal body functions, and this leads to Kwashiorkor. Kwashiorkor is mostly found in countries with a limited food supply. Lack of knowledge on a balanced diet and regional dependence on low protein food can also develop this condition.

**Symptoms**

1. Change in skin and hair colour and texture, Loss of weight, swelling (oedema) of the ankles, feet, and belly, Irritation, compromised immune system, Failure to gain muscle mass, Fatigue, Diarrhoea

**Diagnosis**

The patient is first tested for an enlarged liver or swelling. The level of sugar and protein is measured by simple blood and urine test. Other tests to measure the signs of malnutrition and protein deficiency include:

1. Complete blood count
2. Urinalysis
3. Arterial blood gas
4. Potassium blood levels
5. Creatinine blood levels

**Treatment**

Food with more proteins and more calories can treat Kwashiorkor. Long term vitamin and mineral supplements are advised by the doctors. Calories need to be increased slowly because the patient's diet lacked any significant nutrition for a long period. If there is a delay in the treatment, the child might stay with permanent physical and mental disabilities. If the condition is not treated, it might turn fatal.

***Marasmus:***

There are different types of nutrition deficiency diseases which occur both in children and in adults. This nutrition deficiency disorder is also called as the Malnutrition. The

deficiency diseases are mainly seen in developing countries, areas with high rates of poverty or individuals who have very poor knowledge about the nutrients in food, cleanliness, health, and hygiene.

Our body needs proteins, vitamins, fats, and many other nutrients for the proper functioning of our body cells, tissues, and organs. Among all these nutrients, proteins are called as the building blocks of life as it is the main source of energy and also helps us with the muscle contraction, formation of bones, skin, blood cells, etc.

**Causes**

As mentioned above, it is a nutrition deficiency disease and is mainly caused by:

1. A severe deficiency of protein, vitamins, minerals, carbohydrates, and lipids.
2. Viral, bacterial, and parasitic infections are also a major cause for this disorder.
3. Children, older adults, and people with a weak immune system are more prone to marasmus.

**Other causes**

1. Poverty, Starvation, Famine or unavailability of food, Lack of education about nutritional requirements, Intake of medicines that interfere with the absorption of nutrients.

**Symptoms**

The symptoms of marasmus are more common to the symptoms of kwashiorkor, which includes:

1. Dizziness, Weight loss, Dehydration, Lack of energy, Stunted growth, Chronic Diarrhoea, Shrinkage of Stomach, Respiratory infections, Dry skin and brittle hair, Underweight or loss of muscle mass and subcutaneous fat.

**Diagnosis**

This deficiency disease can be diagnosed by the physician through a physical examination such as:

1. Skin test for the functioning of the immune system.
2. A blood test to check for the presence of any microbial infections.
3. Height, weight and physical activity of a child according to age.
4. General counselling of a child to study the child's mental state and behaviour.
5. Urine and stool test for examining diarrhoea and other nutritional deficiency disorders.

**Treatment**

The first and primary treatment prescribed for the patient suffering from marasmus includes:

1. Multivitamin supplements to improve appetite and nutrition deficiency.
2. Dehydration symptoms can be prevented by the intake of water and other fluids that contain electrolytes.
3. Few antibiotics or other medications are given to treat children with pathogenic infections.
4. A well-balanced diet is given once after the child's condition is stable and starts to recover slowly.
5. Diet rich in protein like skimmed milk, and other energy providing food supplements to increases the energy content.

**Prevention and Safety Measures**

Marasmus is a life-threatening nutrition deficiency disorder which may also lead to death if the symptoms are ignored. Complete recovery can take a month together even with proper diet and medication. Along with the treatment, the patient should also take some preventive steps like:

Drinking boiled water, Eating washed and cooked food, Intake of more water to stay hydrated, Maintain good sanitation and hygiene, Having a healthy and well-balanced diet.

Learning the importance of nutrition and providing good nutrition for both pregnant women, lactating mothers and educating every mother to breastfeed their infants for 5

– 6 months to protect them from nutritional difficulties and to improve their immunity.

**Goiter**

The word goiter indicates an enlarged thyroid. The presence of a goiter does not mean that the thyroid gland is malfunctioning. It could happen to a gland that is generating too less hormone (hypothyroidism) or too much hormone (hyperthyroidism). A goiter indicates a condition that causes the thyroid to grow unevenly.

**Symptoms**

The most common sign of a goiter is a lump in the neck which is benign usually. When it is large enough, one could see and spot the lump clearly. One could also feel tender to touch and feel difficulty while swallowing. Certain goiters are not large and do not show any symptoms.

**Causes**

1. Iodine Deficiency and Autoimmune Disorders
2. Globally, the most recognized cause of thyroid goiter has been due to iodine deficiency, however, it is rare in the US since there is plenty of iodine in drinking water and certain foods, we eat like table salt. Goiters could be caused also by thyroid nodules.
3. Hashimoto's thyroiditis is the next most common cause in countries that use iodized salt. Goiter could also result in cyanide poisoning which is particularly common in tropical countries where people like eating cyanide-rich cassava root in staple food.

**Diagnosing**

The doctor could spot a thyroid goiter in a physical test by examining your neck on whether it is large enough or not. But when the diagnosis has to be confirmed, the doctor might suggest few tests including blood tests, a thyroid, an ultrasound and a radioactive scan.

## Anemia:

Anemia, also spelled as Anemia is the most common blood-related disorder which occurs mainly by the abnormally low levels of red blood cells.

Anemia is also referred to as the iron deficiency disorder because iron is required to produce hemoglobin, a protein in red blood cells that facilitates to carry oxygen all around the body. There are more than 400 types of anemia, which are caused by blood loss, decreased in the production of red blood cells, the destruction of red blood cells, etc.

## Causes

Anaemia is a blood disorder that occurs when our blood does not have enough red blood cells. This deficiency can also be caused by:

1. Autoimmune reaction, Destruction of red blood cells, Severe blood loss by an injury or any accidents, Poor diet and deficiency of iron mineral in the body
2. In very rare cases, anemia can also be hereditary, which affects infants at the time of birth.

## Symptoms

The anaemia symptoms vary according to the type of anaemia, the severity and any other health-related issues including hemorrhaging, ulcers, menstrual problems in women's, cancer, etc.

## Common Symptoms of anemia include:

1. Pica, Insomnia, Seizures, Jaundice, Vomiting, Pale skin, Leg cramps, Constipation, Abdominal pain, Severe joint pains, Shortness of breath, Headache and Dizziness, Susceptibility to infection, Difficulty in concentrating, Unusual or rapid heartbeat.
2. Fatigue and Loss of energy, Stiffness of the arms and legs, Dark Brown or red coloured urine, A tingling sensation in the hands and feet. Delayed growth and development in children.

Anemia is the most common blood disorder which affects 24.8(%) of the world's population. children and women's are at a greater risk of developing this blood disorder and it increases as people grow older. There are many causes of Anemia, certain factors include a hereditary, poor diet, deficiency of certain nutrients, and loss of blood during pregnancy, menstrual cycle, injuries, and any other internal bleedings.

**Scurvy:**

Scurvy is a disease caused by a severe deficiency of vitamin C hence it is also called as the chronic vitamin C deficiency disorder. It is an ancient disease that developed during the 15th to 18th centuries or during the early times of Greek and Egyptians. It is a very rare disease and most commonly seen in undeveloped countries with the lack of proper nutrients.

Vitamin C is a water-soluble vitamin which plays a vital role in repairing, production, regeneration, and maintenance of tissues, bones, blood vessels, and skin. It is a powerful antioxidant, which helps in protecting against heart-related disorders, a variety of cancers and neutralizes the effects of nitrites. Vitamin C also absorbs iron and prevent scurvy, and decrease LDL or bad cholesterol and triglycerides.

**Symptoms**

The deficiency of vitamin C disorders usually appears after 8 to 12 weeks. Fatigue, irritability, lethargy, loss of appetite, low-grade fever, weakness and weight loss are few early signs of Scurvy. Other symptoms include:

1. Nausea, Anemia, Headache, Exhaustion, Bulging eyes, Chest pain, Bleeding gums, Blurred vision, Corkscrew hairs, Sensitivity to light, Shortness of breath, Slow wound healing, Dry and eye irritation, Myalgia, or muscle pain, Painful joints and muscles, Gastrointestinal bleeding, Tender and swollen joints, Depression or mood swings, Tooth decay or loss of tooth, Petechiae or small red spots on the skin.

## Causes

1. The main cause of Scurvy is an insufficient intake of vitamin C or ascorbic acid in our diet.
2. Vitamin C is required for the production of collagen which is an important component in the connective tissue. The connective tissues are required to provide structure and support to the body. It also provides structure to the blood vessels. Lack of vitamin C leads to the underproduction of collagen which results in the breakdown of body tissues.
3. Vitamin C also helps in iron absorption. Deficiency of vitamin C leads to anemia.
4. There are few medical conditions and certain lifestyle changes that result in the deficiency of vitamin C. These include- age factors, consumption of alcohols, use of tobacco products, addiction to drugs, living alone, malnutrition, disabilities, patients suffering from kidney disorders, intake of more carbohydrates, etc.

## Treatment

There are few physical examinations and body fluid analysis to check the levels of vitamin C and iron in the blood serum. X-rays of joints are also used to diagnose this disorder.

The treatment of Scurvy is very natural. This disorder can be cured by taking vitamin C supplements and by increasing the daily intake of vitamin C in the diet. Along with these, patients are also advised to consult a dietitian to alter their diet based upon their medical condition and to reduce the use of drugs, tobacco products and intake of alcohol.

## Sources

As we all are aware of, Vitamins are the essential nutrients which are required by our body cells and tissues to perform several important processes. Vitamin C is rich in all citrus fruits such as oranges, lemons, sweet limes, grapefruits, etc. Other sources include:

1. Vegetables such as bell peppers, broccoli, cabbage, carrots, green peas, tomatoes, potatoes, and, dark leafy green vegetables, especially spinach, kale, and Swiss chard.
2. Fruits including strawberries, blackberries, guava, kiwi fruit, papaya, pineapples, and mangoes.
3. Other are sprouts, paprika, liver, and oysters are also considered as good sources vitamin C.

**Beriberi**

It is a nutritional deficiency disease mainly characterized by low levels of a vitamin B-1 or thiamine in the body cells. Therefore, it is also called as the Thiamine deficiency disorder. Beriberi may be a life-threatening nutritional deficiency disease if it is left untreated.

There are two different types of Beriberi disease:

1. **Wet beriberi –** This disorder usually affects the heart and other organs of the circulatory system. In extreme cases, this may also lead to a heart attack and other heart disorders.
2. **Dry beriberi –** This disorder disturbs the functioning of the nerves of the nervous system and leads to damage or decreased muscle strength, which eventually results in paralysis of muscles.

Food rich in thiamine reduces the risk of being affected with beriberi. The disease can be seen in women with extreme nausea, vomiting in pregnancy, people suffering from AIDS, etc.

**Symptoms:**

***Wet beriberi* -** Increased heart rate, Swelling of the lower legs, Shortness of breath with activity, MOTN – Middle-of-the-night insomnia

***Dry beriberi* -** Pain, Tingling, Vomiting, Confusions, Difficulty in walking, Strange eye movements, Difficulties in speech, Loss of sensation in hands and feet, Loss of muscle functioning or paralysis of the lower limbs**.**

Beriberi is also associated with Wernicke-Korsakoff syndrome. This is a form of brain damage caused due to thiamine deficiency. Wernicke encephalopathy causes damage to the thalamus and hypothalamus of the brain.

This causes symptoms such as: confusion, loss of memory, muscle coordination loss, problems in vision Korsakoff syndrome damages the memory-forming region of the brain.

This leads to symptoms such as:

1. memory loss
2. hallucinations
3. problems in forming new memories

**Causes**

Beriberi is a very rare disease, which is caused by the nutritional deficiency of vitamin B1 or thiamine. According to recent research, this disorder is very rare in the United States. This is because all the food is vitamin enriched nowadays.

Beriberi is also seen in most of the people who are alcohol abuse, as alcohol leads to poor nutrition and weakens the body cells to absorb and store vitamin B1 from the food we eat.

In very rare cases, beriberi can also be a genetic disorder, transmitted from the mother to their infants during breastfeeding only in those case if the mother's body is in lack of thiamine.

The infants who are breastfed by a mother suffering from Vitamin B1 deficiency also suffer from this disease. Also, dialysis and heavy doses of water pills increase the risk of beriberi.

**Diagnosis**

There are a series of physical examination and other medical tests used to determine whether a person is suffering from beriberi. The test includes:

1. Certain breathing and lung functioning tests.

2. Blood and urine tests to check the levels of thiamine in the blood cells.
3. A physical examination of the legs to find if any swelling in both the lower legs.
4. Examination of the heart to check the variations in the heartbeat and the enlargement of the heart.
5. Neurological examination to check for the changes in the walking, coordination problems, decreased reflexes, etc.

**Treatment**

The treatment of Beriberi is very natural. This disorder can be cured by replacing the thiamine level in the blood cells by increasing the daily intake of vitamins in the diet. Patients are treated with the Thiamine supplements, which are given orally or through a shot (injection). Along with the diet and medications, patients are also advised with follow-up blood tests to monitor how well the body is absorbing the vitamin.

**Prevention:**

To prevent beriberi, maintaining a healthy and balanced diet is important. Include foods rich in thiamine and if a patient is alcohol abuse, then limiting alcohol consumption or completely stop or quit will reduce your risk of developing beriberi.

Foods that are rich in vitamin B1 or thiamine are – beans, cereals, milk and milk products, eggs, fish, legumes, meat, nuts, seeds, whole grains, and certain vegetables, including acorn, asparagus, beet greens, brussels sprouts, green leafy vegetables, squash, and tomatoes.

**Rickets:**

Rickets is a childhood disorder of the bones where the bones become soft and prone to fractures. Bones tend to become weak due to an inadequate supply of nutrients, Vitamin D3 in particular. It is also caused due to lack of calcium and phosphate in the body. Weak bones can lead to bone deformities. Vitamin D largely comes from exposing the skin to sunlight. Other good sources of Vitamin D are fatty fish (such as tuna and salmon) and egg yolks.

It is a commonly occurring disorder in children, especially ones with darker skin due to lack of exposure to sunlight, is also seen in premature infants. This condition of rickets in adults is called osteomalacia which is usually characterized by soft bones.

Diagnosis of rickets is normally done through blood tests and X-rays. Blood tests indicate the condition through alarmingly low levels of calcium and phosphorous and higher levels of a phosphatase that is alkaline in nature. Through X-rays, it is indicated by the change in the shape of bones. Bone biopsies also help in their diagnosis.

**Types**

Rickets is of the following types:

1. Nutritional Rickets– This is caused due to intake of food which lacks in calcium, phosphorous and vitamin D.
2. Hypophosphatemia Rickets- It is caused due to low levels of phosphate. It is an X-linked genetic disorder where the kidneys are not able to control the amount of phosphate excreted in the urine.
3. Renal Rickets- People suffering from kidney disorders have renal rickets. They are unable to regulate the amount of calcium and phosphate excreted in the urine.

**Symptoms**

The symptoms of rickets are as follows:

1. Reduced growth and short height
2. Fractures in bones
3. Softening of bones
4. Pain in the bones of arms, legs, pelvis, and spine
5. Deformities in teeth
6. Deformities in the skeleton like bowlegs, and protruding breastbone.

**Causes**

Vitamin D is required to absorb calcium from food. Lack of calcium and vitamin D or inability to absorb the same causes rickets.

Vitamin D is obtained through sunlight and food. Vitamin D is produced when the skin is exposed to sunlight. Use of sunscreen blocks the rays and hence the production of Vitamin D by the skin is reduced.

Fish oil and fat and the egg yolk are highly rich in Vitamin D. In addition, some cereals, milk and fruit juices also contain Vitamin D.

Sometimes, Rickets can be inherited as it is a genetic disorder. It commonly occurs in children since bones grow rapidly in them, more so in children with the following attributes:

1. Darker complexion – excess melanin pigment
2. Inadequate supply of foods rich in calcium, phosphorous
3. Lack of sunlight exposure
4. When infants are breastfed without Vitamin D supplements
5. Possessing diseases which inhibit the absorption of Vitamin D

**Treatment**

This condition can be treated under proper guidance if the underlying cause is diagnosed early on. It can also be treated without any major bone deformities; however, certain conditions might need surgical intervention.

1. Rickets can be treated by eating a diet rich in vitamins and minerals, especially vitamin D, consuming calcium and phosphorus rich food.
2. Vitamin D supplements need to be given to infants of the breastfeeding age. Enough exposure of the skin to the sunlight improves the condition.
3. If rickets is caused due to a genetic disorder, the patient is prescribed vitamin D hormones and phosphorous medications.
4. Braces might be required in case of skeletal deformities to position the bones correctly as the child grows.

**Genetic Disorders**

A genetic disorder is a disease or syndrome that is caused by an abnormality in an individual DNA. Abnormalities can range from a small mutation in a single gene to the addition or subtraction of an entire chromosome or even a set of chromosomes.

Genetic disorders are of two types namely, Mendelian disorders and chromosomal disorders.

**Mendelian disorders:**

Alteration or mutation in a single gene causes Mendelian disorders. These disorders are transmitted to the offsprings on the same line as the Mendelian pattern of inheritance. Some examples for Mendelian disorders are Thalassemia, albinism, phenylketonuria, sickle cell anemia, Huntington's chorea, etc., These disorders may be dominant or recessive and autosomal or sex linked.

**Thalassemia**

Thalassemia is an autosomal recessive disorder. It is caused by gene mutation resulting in excessive destruction of RBC's due to the formation of abnormal hemoglobin molecules. Normally hemoglobin is composed of four polypeptide chains, two **alpha** and two **beta** globin chains. Thalassemia patients have defects in either the alpha or beta globin chain causing the production of abnormal hemoglobin molecules resulting in anemia. Thalassemia is classified into alpha and beta based on which chain of hemoglobin molecule is affected. It is controlled by two closely linked genes HBA1 and HBA2 on chromosome 16. Mutation or deletion of one or more of the four alpha gene alleles causes **Alpha Thalassemia**. In **Beta Thalassemia,** production of beta globin chain is affected. It is controlled by a single gene (HBB) on chromosome 11. It is the most common type of Thalassemia and is also known as Cooley's anemia. In this disorder the alpha chain production is increased and damages the membranes of RBC.

**Phenylketonuria**

It is an inborn error of **Phenylalanine** metabolism caused due to a pair of autosomal recessive genes. It is caused due to mutation in the gene PAH (phenylalanine hydroxylase gene) located on chromosome 12 for the hepatic enzyme "phenylalanine hydroxylase" This enzyme is essential for the conversion of phenylalanine to tyrosine. Affected individual lacks this enzyme, so phenylalanine accumulates and gets converted to phenyl pyruvic acid and other derivatives. It is characterized by severe mental retardation, light pigmentation of skin and hair. Phenyl pyruvic acid is excreted in the urine.

## Albinism

Albinism is an inborn error of metabolism, caused due to an autosomal recessive gene. Melanin pigment is responsible for skin colour. Absence of melanin results in a condition called albinism. A person with the recessive allele lacks the tyrosinase enzyme system, which is required for the conversion of dihydroxyphenyl alanine (DOPA) into melanin pigment inside the melanocytes. In an albino, melanocytes are present in normal numbers in their skin, hair, iris, etc., but lack melanin pigment.

## Huntington's chorea

It is inherited as an autosomal dominant lethal gene in man. It is characterized by involuntary jerking of the body and progressive degeneration of the nervous system, accompanied by gradual mental and physical deterioration. The patients with this disease usually die between the age of 35 and 40.

## Chromosomal Abnormalities

Each human diploid (2n) body cell has 46 chromosomes (23 pairs). Chromosomal disorders are caused by errors in the number or structure of chromosomes. Chromosomal anomalies usually occur when there is an error in cell division. Failure of chromatids to segregate during cell division resulting in the gain or loss of one or more chromosomes is called aneuploidy.

It is caused by nondisjunction of chromosomes. Group of signs and symptoms that occur together and characterize a particular abnormality is called a syndrome. In humans, Down's syndrome, Turner's syndrome, Klinefelter's syndrome, Patau's syndrome are some of the examples of chromosomal disorders.

### a. Autosomal aneuploidy in human beings

Several autosomal aneuploidies have been reported in human being e.g. Down's syndrome (21-Trisomy), Patau's syndrome (13-Trisomy).

### 1. Down's Syndrome/Trisomy – 21

Trisomic condition of chromosome – 21 results in Down's syndrome. It is characterized by severe mental retardation, defective development of the central nervous system,

increased separation between the eyes, flattened nose, ears are malformed, mouth is constantly open and the tongue protrudes.

**2. Patau's Syndrome/Trisomy-13**

Trisomic condition of chromosome 13 results in Patau's syndrome. Meiotic non disjunction is thought to be the cause for this chromosomal abnormality. It is characterized by multiple and severe body malformations as well as profound mental deficiency. Small head with small eyes, clef palate, malformation of the brain and internal organs are some of the symptoms of this syndrome.

**b. Allosomal abnormalities in human beings**

Mitotic or meiotic non-disjunction of sex chromosomes causes allosomal abnormalities. Several sex chromosomal abnormalities have been detected. Eg. Klinefelter's syndrome and Turner's syndrome.

**1. Klinefelter's Syndrome (XXY Males)**

This genetic disorder is due to the presence of an additional copy of the X chromosome resulting in a karyotype of 47, XXY. Persons with this syndrome have 47 chromosomes (44AA+XXY). They are usually sterile males, tall, obese, with long limbs, high pitched voice, under developed genitalia and have feeble breast (gynecomastia) development.

**2. Turner's Syndrome (XO Females)**

This genetic disorder is due to the loss of a X chromosome resulting in a karyotype of 45, X. Persons with this syndrome have 45 chromosomes (44 autosomes and one X chromosome) (44AA+XO) and are sterile females. Low stature, webbed neck, under developed breast, rudimentary gonads lack of menstrual cycle during puberty, are the main symptoms of this syndrome.

**Questions:**

1. Define Health? What are the signs of good health?
2. Discuss about Personal and Community Health.
3. Mention some of the Human diseases with its symptoms, treatment, and preventive measures.